(1) Your name and student ID

- 113062535 李佳翎

(2) How to compile and execute your program and give an execution example.

先去 HW3/src,打 make

然後去 HW3/bin

./bin/hw3 <input.txt> <output.out> dead_space_ratio

./bin/hw3 ../testcase/public1.txt ../output/public1.out 0.1

(3) The wirelength and the runtime of each testcase with the dead space ratios 0.15 and 0.1, respectively. Paste the screenshot of the result of running the HW3_grading.sh.

```
HW3/bin/hw3
+-------------------------------------------------+
|                                                 |
|    This script is used for PDA HW3 grading.     |
|                                                 |
+-------------------------------------------------+
host name: ic21
compiler version: g++ (GCC) 9.3.0

grading on 113062535:
 checking item          | status
------------------------|--------
 correct tar.gz         | yes
 correct file structure | yes
 have README            | yes
 have Makefile          | yes
 correct make clean     | yes
 correct make           | yes

  testcase |      ratio | wirelength |    runtime | status
-----------|------------|------------|------------|--------
   public1 |       0.15 |     209625 |      99.77 | success
   public2 |       0.15 |     440734 |     592.57 | success
   public3 |       0.15 |     559636 |     592.69 | success
   public1 |        0.1 |     216859 |     114.05 | success
   public2 |        0.1 |     460701 |     592.95 | success
   public3 |        0.1 |     567218 |     592.96 | success
+-------------------------------------------------+
|                                                 |
|   Successfully write grades to HW3_grade.csv    |
|                                                 |
+-------------------------------------------------+
```

(4) Please show that how small the dead space ratio could be for your program to produce a legal result in 10 minutes.

	Public1	Public2`	Public3
dead space ratio	0.05	0.08	0.05

(5) The details of your implementation. If there is anything different between your implementation and the algorithm in the DAC-86 paper, please reveal the difference(s) and explain the reasons.

- Initial floorplan->If the width or height are bigger than the outline, do SA1(for minimizing area) , then do SA2(for minimizing length).

要做 SA1 的原因是，因為 public2 無論如何 initial 都放不進去，如果同時優化 area 跟 wirelength 的話，會花超過 10 分鐘去找答案，所以想出這個兩階段的優化，但如果開始就在 outline 裡面的話，就直接優化 wirelength，而我的結果來看，如果都在 outline 裡面只要 swap operand 就可以了，其他的改結構感覺很容易超出邊界。

另外還有一個 M4，是 random swap two operand，其實是一開始寫錯，但後來拿來測試發現效果蠻好，於是就保留繼續用。

(6) Please describe your method of your initial floorplan.

我就是先把每個按照 max(height, width)排列，從最大的開始排，如果超過 outline 就往上疊，同時在排的時候也直接考慮旋轉，因此我的 public1 和 public3 一開始都可以直接排進 outline 裡。

舉例來說，假設大小是 1234...，我就是 1234...如果超過 outline，就補 v，變成 1234VVVH 然後繼續從新的一排往右。

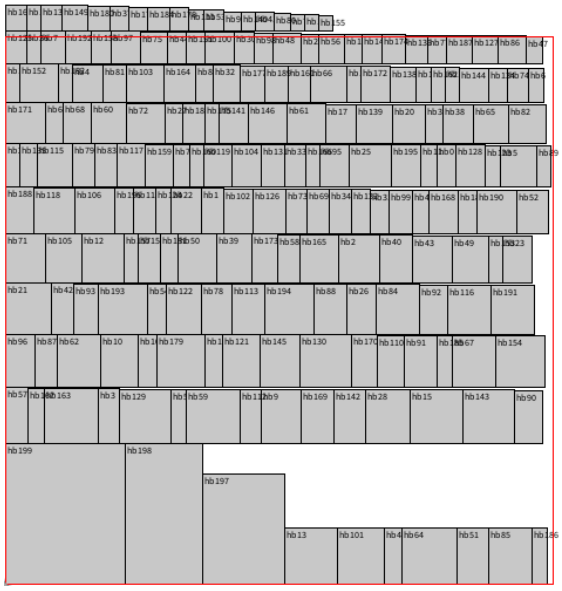
(7) What tricks did you do to speed up your program or to enhance your solution quality? Please use HW3_printer to generate the image to compare different stages (e.g., initial floorplan → trick 1 → trick 2 → final result) of your floorplanning results.

就如上面提到了有新增，m4 這個 move，又增加了更多隨機性。

然後因為我是每次都重新建 tree，所以 code 都會跑超久><。

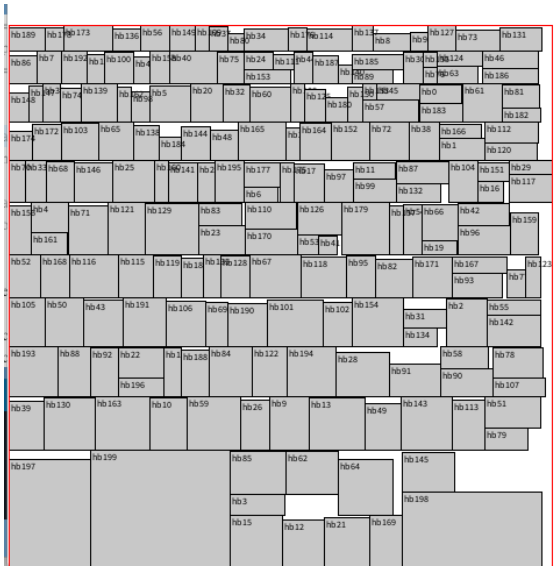
以 public2 dead_space_ratio0.1 為例

Initial floorplan



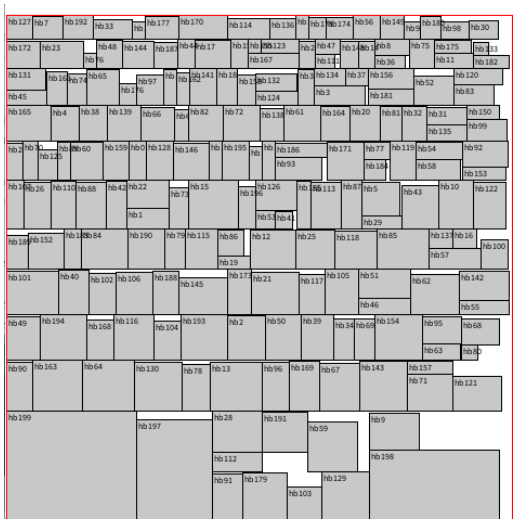
可以看到有些超出 outline 了->

SA1(for area)



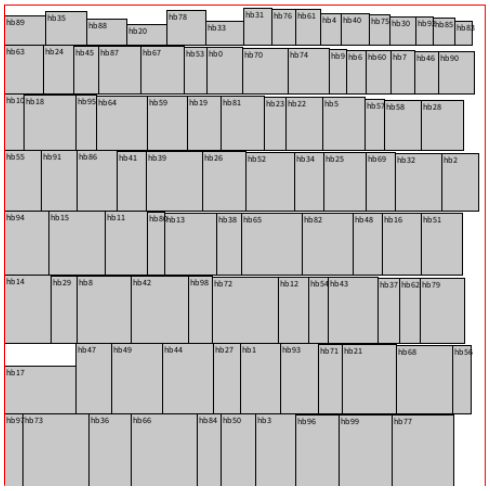
成功塞進 outline，但 wirelength 很差->

SA2(wirelength)

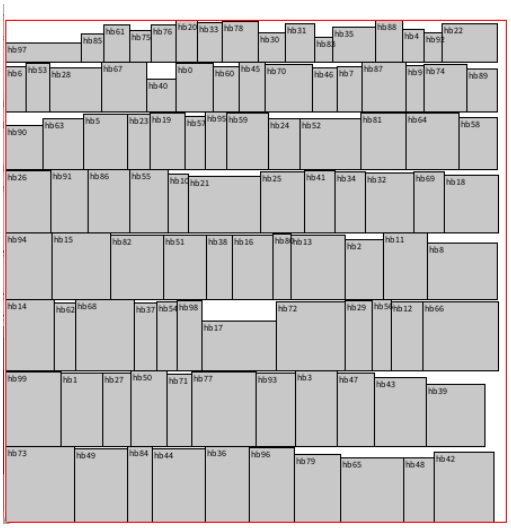


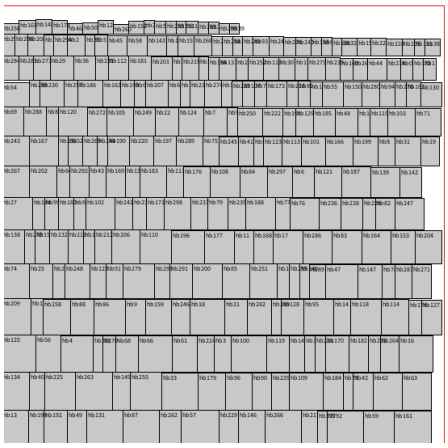
在 outline 內，且 wirelength 有提升。

Public1、public3 都是 initial 完直接做 SA2。如下圖

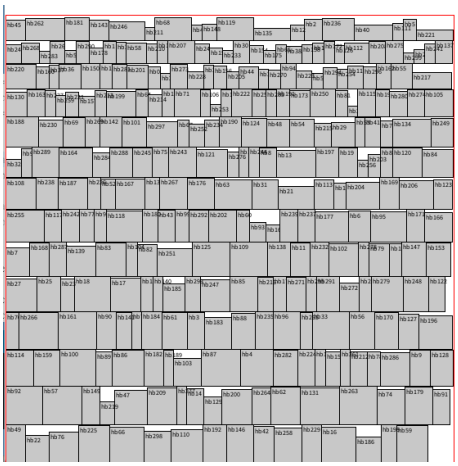


->





->

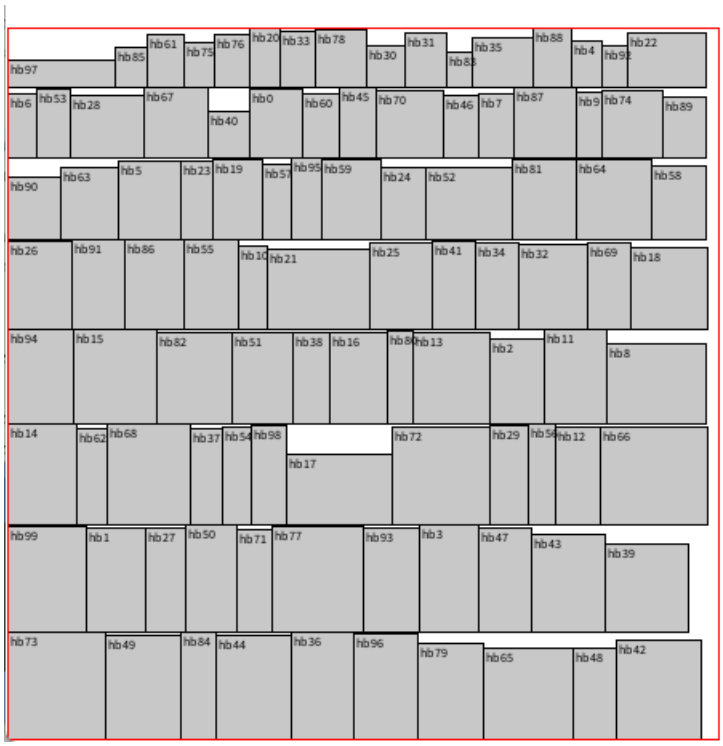


(8) What have you learned from this homework? What problem(s) have you encountered in this homework?

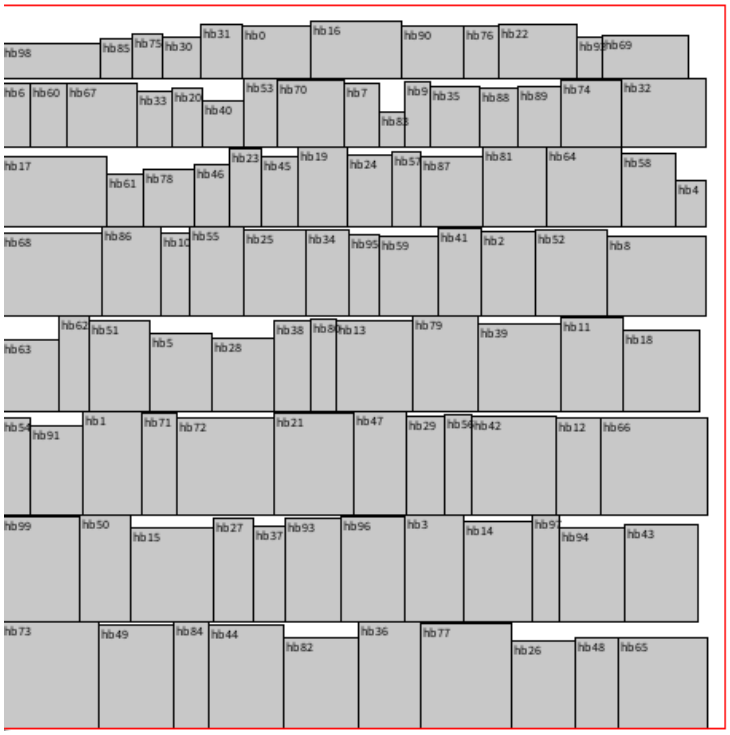
其實這次寫的 code 好像沒有上次那麼難，但如果要寫到優化的部分，就是老師上課提到，每次交換的話只有兩條 path 會動到，這個部份我花了很多時間寫，畢竟指標就是 C++裡面最難的部分，再來其他部分就是要一直調參數，有點在靠運氣 XD。

Result

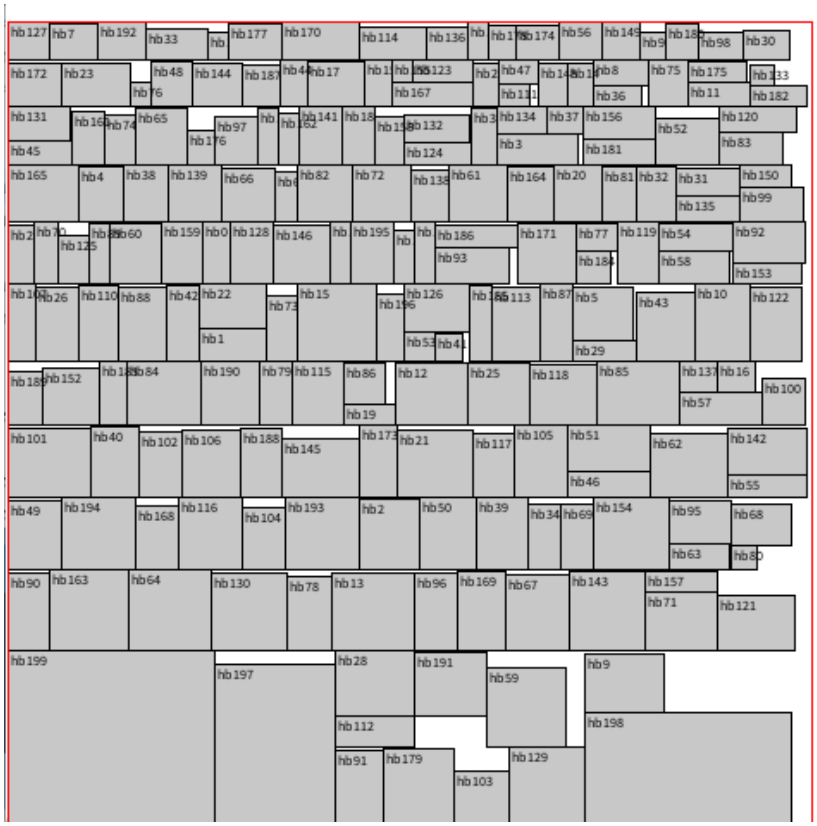
Public1 0.1



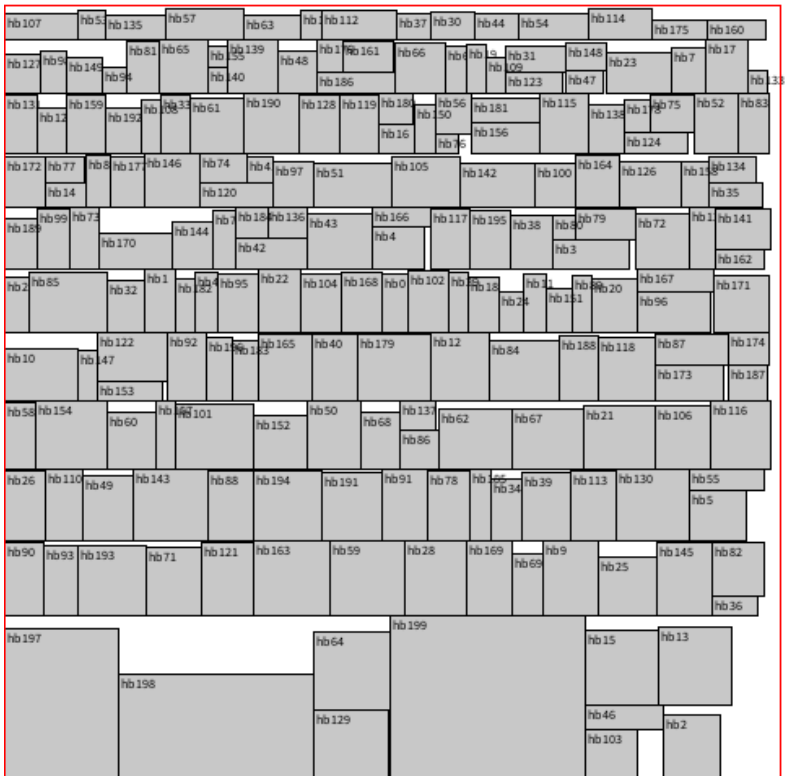
Public1 0.15



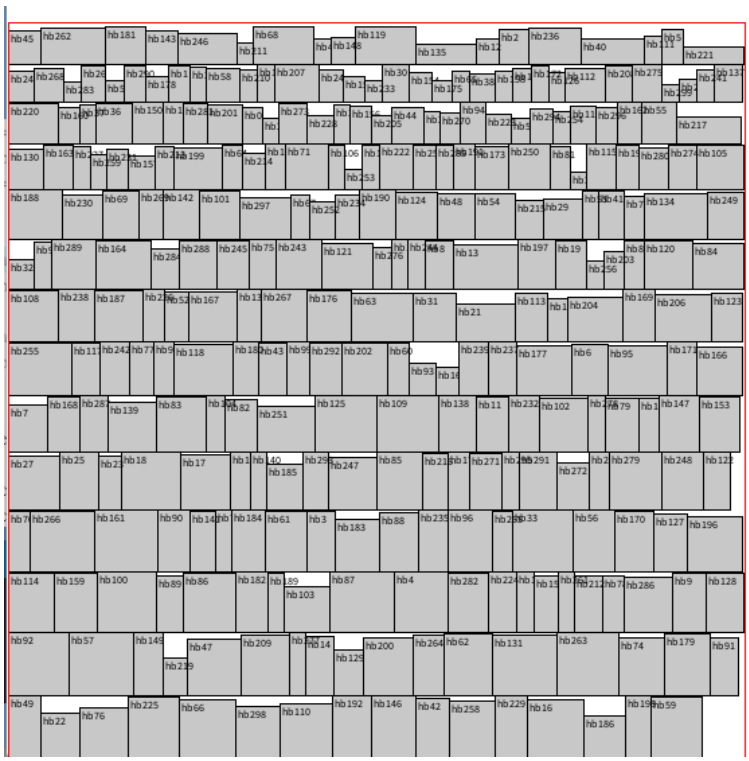
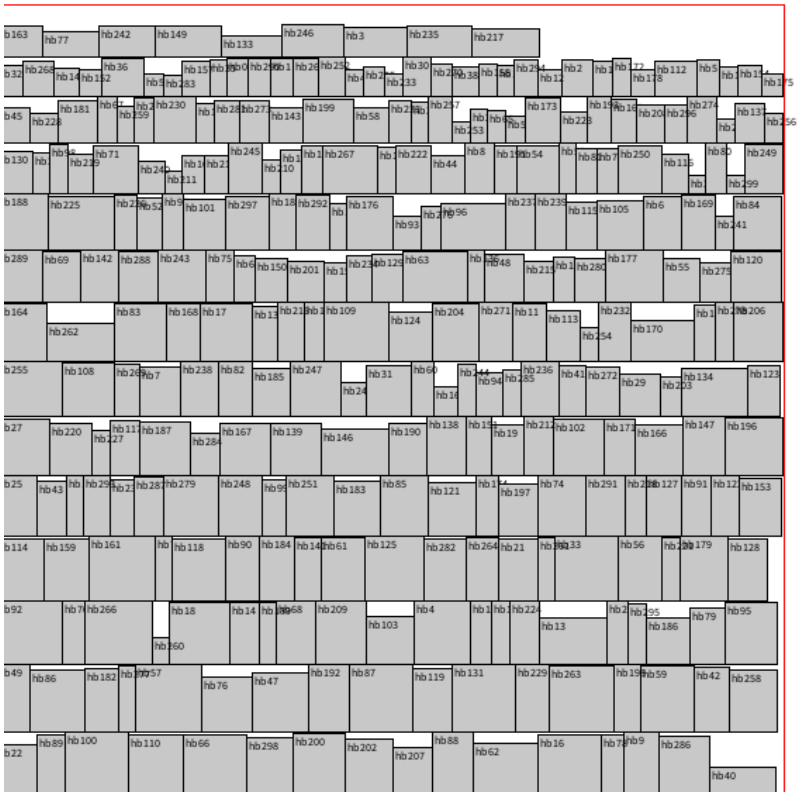
Public2 0.1



public2 0.15



Public3 0.1



public3 0.15